

CS351 - IT Workshop II (Cloud Computing)

Assignment 3

Total Marks: 25

Please follow the instructions below. Marks will be deducted if these instructions are not followed.

(i) **Screenshots:** Take clear screenshots, resize the images to reduce their file size, and upload them to Google Classroom. Please do not upload a .zip file in Google Classroom.

(ii) **Source Code File Naming:** Name each Python source file using the following format: <assignment_no>_<question_no>.py

For Example: a1_q1.py

(iii) **Source Code Header:** Include your name and roll number at the top of each source file using the following format.

Name: <Your Name>

Roll No: <Your Roll No>

(iv) **Proper Comments:** You need to write comments for each block of code, function inputs and outputs, and the working logic of the functions.

(v) **Proper Variable and Function Name:** Please use meaningful variable and function names.

1. Create a **static website** with one or two HTML pages and copy the pages and dependent files in an **Amazon S3 bucket**. **[5 Marks]**
2. Create a **startup script** to install the **Apache server** on an EC2 instance. The script should also copy the website files (from the previous step) from **S3** to the **EC2** instance. **[3 Marks]**
3. Create a Python program using **boto3** to launch a **t3.micro** instance in EC2. Supply the **startup script** you created in the previous step while launching a new instance from this program. **[7 Marks]**
4. Use a **security group** with port **80** open. **[2 Marks]**

5. Your program should check the **status** of the newly launched instance and return the **public DNS address** of the instance when the status changes to **“running”**.

[6 Marks]

6. Open the **public DNS** of the newly launched instance in a browser and verify if the **static website** works.

[2 Marks]

Please note that you need to install **boto3** on your Laptop. You can install **boto3** from source or using the **pip3** command. For installation and **boto3** configuration, you can follow the links below.

<https://boto3.amazonaws.com/v1/documentation/api/latest/index.html>